Final Report

IAS-NASI-INSA Summer Research
Fellowship Programme 2021

Introduction to Particle Physics

under the mentorship
of

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Acknowledgements

I would like to extend my heartiest gratitude to respected Dr. Lokesh Kumar, Dept. of Physics, Panjab University, Chandigarh, for providing me this amazing opportunity to work in my field of interest. His constant guidance and encouraging attitude kept me motivated at all the stages of the project. Also, I would like to thank the Indian Academy of Sciences for facilitating my summer research internship at Panjab University, Chandigarh.

I would also like to thank him for curating the layout of the project in such an innovative way, all the way from basics to performing real-life experiments in Particle Physics. This methodology greatly enhanced the learning as compared to the traditional approaches
.

Although I may not have been at my best productivity, this project not only enabled me to strengthen my basic knowledge of Particle Physics but also provided great insight into the software, resources, and other recent developments in High Energy Physics. I am indebted to Dr. Kumar for providing such a fruitful and enriching learning experience.

I would also like to sincerely acknowledge the resources and tools provided by the CERN CMS Masterclass that helped me complete this project. The online event viewer helped greatly in analyzing and classifying the events. I would also like to express my gratitude to the CERN for providing the measurements for analysis.

Finally, I would like to express my regards to my parents and friends for helping me with this project.

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September 28, 2021

Abstract

This project can be considered a three-stage process. In the first stage, we focused on building our basics of Particle Physics, wherein we studied about Fundamental particles and interactions, antiparticles, Klein Gordon Equation, Symmetry, and Conservation Laws.

Having gained some basic knowledge about Particle Physics, we started to explore CERN Root - a data analysis framework used by Particle Physicists. We learned to plot histograms, visualize functions, plotting data from files, and using TTree for huge datasets.

The third stage dealt with using iSpy event viewer by CERN to display the measurements taken in CERN, and to classify their likelihood of being a J/ψ event by analyzing the dimuon tracks.

In essence, we started from the basics of Particle Physics and then tried to connect that with the recent experiments being conducted in Particle Physics.

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1 Introduction to Particle Physics

Particle Physics is the study of fundamental constituents of matter and their interactions. With the advent of technology, the particles that are regarded as 'fundamental' have changed.

1.1 The Standard Model

Figure 1: The Standard Model of particle Physics
Figure taken from [1]

The Standard Model of particle physics deals with four kinds of particles,

1. Spin 1/2 particles: Quarks, Fractional Charge. There are in all six quarks, u(up), d(down), s(strange), c(charm), b(bottom) and t(top). The u,c,t have a charge $\frac{2e}{3}$ and d,s,b have $-\frac{e}{3}$.

2. Spin 1/2 particles: Leptons, Integral charge. There are six different flavors of- electron(e), muon(μ), taon(tau) and their corresponding neutrinos- electron type neutrino ν_e, muon type neutrino ν_μ, taon type neutrino ν_τ. Neutrinos are charge neutral whereas electron, muon and taon have charge -e.

3. Spin 1 particles: Gauze bosons, act as force carriers.

4. Spin 0 particles: Higgs Bosons, the origin of mass.

The leptons exist as free particles, whereas the quarks exist as baryon(three quarks) and mesons(quark-antiquark) pair. Baryons and mesons are collectively known as *hadrons*.

The Standard Model is a *renormalizable* field theory and is free from 'anomalies' as the total charge of all the fermions is 0.

1.2 Fundamental Interactions

Quantum Mechanically, the interactions between the fundamental particles are defined in terms of the exchange of bosons between the particles rather than via classical fields.

Four types of fundamental interactions are described by the Standard Model,

Strong Interactions: They are the forces that bind the quarks in neutrons and protons. They also bind the neutron and protons inside the nucleus. These forces are mediated by the gluon(G).

Weak Interactions: is produced in the nuclear β-decay, leading to the emission of electron and neutrino. $W^{\pm}$ and Z^0 bosons mediate the weak interactions.

Electromagnetic Interactions: They are responsible for most of the forces occurring beyond the nuclei and also play an important role in intermolecular forces in solids and liquids. They are mediated via the photons(γ).

Gravitational interactions They act between all the particles. However, in the microscopic regime of particle physics, they are very weak. They are hypothesized to be mediated by a boson known as *graviton*.

The relative strengths of the interactions can be described as Strong > Electromagnetic >> weak >>> Gravitational

Interaction	Mediator	Spin
Strong	gluon, G	$1^{-}1$
Weak	$W^{\pm}$, Z^0	$1^{-}1$
Electromagnetic	photon,γ	$1^{-1}, 1^{+1}$
Gravitational	gravion,g	2^{+1}

Table 1: Fundamental Interaction Mediators

1.3 Antiparticles

The antiparticles are objects with the same mass and lifetime as the particle but having opposite signs of charge and magnetic moment. This concept was first formulated for fermions by Dirac in 1928.
The idea of antiparticles originated from the equation,

$$E^2 = p^2c^2 + m^2c^4 \tag{1}$$

$$T = E - mc^2 = mc^2(1 + p^2/m^2c^2)^{\frac{1}{2}} - mc^2 \approx p^2/2m \tag{2}$$

$$E = \pm\sqrt{p^2c^2 + m^2c^4} \tag{3}$$

$$\tag{4}$$

Thus, equation (3) shows that theoretically, it is possible to have negative energies. Quantum Mechanically, such particles with negative energies can be explained as traveling back in time towards the negative x-direction. Both the fermions and leptons have antiparticles, but the conservation law holds only for fermions, stating that fermions-antifermions can be created only in pairs.

1.4 Klein Gordon Equation

The relativistic wave equation or Klein Gordon equation is,

$$\frac{\partial^2 \Psi}{\partial t^2} = \nabla^2 \Psi - \frac{m^2 c^2}{\hbar^2} \Psi \tag{5}$$

The Klein Gordon equation is used to define scalar(spinless) particles. It is second order in space and time. The Weyl equations (6) ensure that physical processes stay relativistically invariant, i.e., they are first derivative in both time and space.

$$\frac{\partial \Psi}{\partial t} = \pm\Big(\sigma_1 \frac{\partial \Psi}{\partial x} + \sigma_2 \frac{\partial \Psi}{\partial y} + \sigma_3 \frac{\partial \Psi}{\partial z}\Big) \tag{6}$$

The famous *Dirac Equation* can be derived by Using the quantum mechanical operators for energy and momentum and including the mass of fermions.

$$E\Psi = (\alpha.\vec{p} + \beta m)\Psi \tag{7}$$

α and β are 4x4 matrices.

1.5 Yukawa Potential

Yukawa, in 1935 postulated that the short-range forces inside the nucleus are the result of interactions due to massive quanta exchange.
The Klein Gordon equation (5) transforms to wave equation for propagation of EM wave if we set m = 0.Ψ can be viewed as Potential at a point $U(\vec{r})$

$$\nabla^2 U(r) = \frac{1}{r^2}\frac{\partial}{\partial r}\Big(r^2 \frac{\partial U}{\partial r}\Big) = \frac{m^2 c^2}{\hbar^2} U(r) \tag{8}$$

Integrating to find the potential as a function of r, we get

$$U(r) = \frac{g_0}{4\pi r} e^{-\frac{rmc}{\hbar}} \tag{9}$$

Equation (9) is The Yukawa potential, a simple relationship between the range of field and the mass of interacting quantum. g_0, a constant of integration can be viewed as *strong nuclear charge.*

In QFT, the propagator is a function that represents the probability amplitude for a particle to travel from one place to another in a given time, having certain energy and momentum. The scattering amplitude for a single boson exchange process is

$$f(q^2) = \frac{g_0 g}{q^2 + m^2} \tag{10}$$

1.6 Feynman Diagrams

Feynman diagrams[2] provide us a graphical way to display and understand the interactions between particles and fields. There is an elaborate algebra that is used to compute the matrix elements corresponding to the interactions[3].

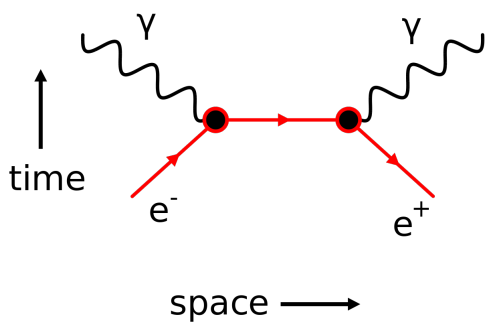


Figure 2: Feynman Diagram for electron-positron annihilation
Figure taken from [3]

As a matter of convention, the Feynman Diagrams follow some rules,

- Fermions are represented by solid straight lines while the bosons(interaction mediators) are represented using the wavy curly lines.
- The arrows along the solid lines indicate the time. Time increases from left to right.
- The junction of two lines represents the interaction energy and momentum are conserved here.
- Real particles are represented by lines entering/leaving at the boundary of the figure. The lines connecting the vertices represent virtual state.

1.7 Interaction Cross Section(σ)

The interaction cross section measures the likelihood of interaction between two particles. It is related to the reaction rate per target particle(W) and total incident flux(ϕ) by

$$\sigma = W/\phi \tag{11}$$

According to Non Relativistic perturbation theory,

$$W = \frac{2\pi}{\hbar}\left|M_{if}\right|^2 \tag{12}$$

$$M_{if} = \int \Psi_f^* U \Psi_i dV$$

Ignoring the effects of the spin, for a reaction a + b $\longrightarrow$ c + d, we have

$$\frac{d\sigma}{d\Omega} = \frac{1}{\pi^2\hbar^4}\left|M_{if}\right|^2 \frac{p_f^2}{\nu_i\nu_f} \tag{13}$$

ν_i, ν_f are the relative velocities of a and b in Centre of Mass frame.

1.8 Decays and Resonances

The mean lifetime of a decay state is represented $\tau = 1/W$, where W is the reaction rate (11). However, for strong decays, one uses natural width(Γ) instead of τ.

$$\Gamma = \frac{\hbar}{\tau} = \hbar W = 2\pi M^2 \int \rho_f d_\Omega \tag{14}$$

The decay rate for a particle P will be the fractional number of decays of a group of P particles, per unit time. The exponential decay rate is

$$N_P(t) = N_P(0)exp\big(-\frac{\Gamma t}{\hbar}\big) \tag{15}$$

Wavefunction representing a non-stationary decay state having central frequency($\omega_R = E_R/\hbar$), is

$$\psi(t) = \psi(0)e^{t(iE_R+\Gamma/2)} \tag{16}$$

E_R is the resonance energy.
The amplitude as a function of energy is

$$\chi(E) = \int \psi(t)e^{\iota Et}dt = \Psi(0)\int \psi(t)e^{t[(\Gamma/2)+\iota(E_R-E)]}dt = \frac{K}{(E-E_R)-\iota\Gamma/2} \tag{17}$$

For some constant K.
The cross section $\sigma(E)$ representing the probability of two particles to interact and for a third particle is

$$\sigma(E) \propto \chi(E)\chi^*(E)$$

Thus,

$$\sigma(E) = \sigma_{max}\frac{\Gamma^2/4}{(E-E_R)^2+\Gamma^2/4} \tag{18}$$

We also include the spin multiplicity factor (2s +1) to obtain the complete expression for cross section, Breit Wigner Formula (19)

$$\sigma(E) = \frac{\sigma_max}{(2s_a+1)(2s_b+1)}\frac{\Gamma^2/4}{(E-E_R)^2+\Gamma^2/4} \tag{19}$$

1.9 Isospin(I)

Isospin(I): Isospin is a quantum number that is used to group particles having similar mass, for example, proton and electron [4]. The proton is said to have *isospin projection* of +1/2 whereas the electron has -1/2. Isospin indicates that strong interactions are charge independent.

The Isospin, Strangeness(S) and Baryon(B) are not independent quantum numbers. They are related as

$$\frac{q}{|e|} = I_3 + \frac{S+B}{2} = I_3 + \frac{Y}{2} \tag{20}$$

Y is the Hypercharge($=S+B$)
S: Strangeness
B: Baryon Number
I_3 : Isospin Projection Quantum Number
Equation (20) is also known as the Gell-Mann-Nishijima formula. However, in the case of the elementary nucleon reactions, the interquark potential is more complex than Yukawa.

1.10 Conservation Laws

Through experimental evidence, it was witnessed that certain types of interaction are more common, and certain are forbidden. This led to the formulation of the Conservation Laws. These Conservation Laws [5] are in addition to the Basic conservation of energy, momentum, angular momentum.

The three basic symmetries that usually(not always) hold are [6] ,

1. Charge Conjugation(C): Reversing the sign of electric charge and that of all the internal quantum numbers.

2. Parity(P): Spatially inverting the coordinates but not the time.

$$P\Psi(\vec{r}) = P\Psi(\vec{-r})$$

3. Time Reversal(T): *Reversing the time* by replacing t with -t. This operation reverses the time derivatives like angular momentum.

Conserved quantity	Strong	Interaction Type Electromagnetic	Weak
Energy-Momentum charge baryon number lepton number	yes	yes	yes
CPT	yes	yes	yes
P(Parity)	yes	yes	no
C(Charge Conjugation Parity)	yes	yes	no
CP(or T)	yes	yes	$10^{-}3$ violation
I(Isospin)	yes	no	no

Table 2: Conservation Rules for the Fundamental Interactions [2]

1.11 OZI rule

The OZI rule is invoked to explain why certain decay modes appear less frequently than otherwise expected. It states that any process occurring due to strong forces will be suppressed if its Feynman Diagram can be separated into two disconnected diagrams-one containing all the initial states and the other containing all the particles in the final state, only through removal of internal gluon lines.

OZI rule is a consequence of quantum electrodynamics as with the increase in the energy and momentum transfer, the coupling constant decreases.

2 CERN Root

Root is a standard scientific software toolkit. It has all the components needed to deal with big data processing, statistical analysis, visualization and storage. It is primarily written in C++ but integrated with other languages such as Python and R [7]. For this project, the version ROOT v6.08/22 has been used.

We can use root as a simple calculator to perform day-to-day calculations, as a plotter to plot various functions, for data analysis to plot histograms from a data file, to do error analysis on experimental data by plotting error bars, learning C++ [8] using root prompt .

2.1 Plotting a Histogram using data from a file

In this part, we will learn how to use root to plot a histogram, by inputting the data from a .txt file, this type of tasks are frequently encountered in Particle Physics to plot nuclei counts, plotting the muon mass in J/Ψ decay etc.

```
void histogram()
{
  // Creating an object named hist using the class TH1F, this class
     is used to plot histograms.
```

```
  // Syntax: TH1F("object name", "title", bin nos, min_range,
    max_range)
  TH1F  *hist = new TH1F("hist", "Histogram", 7, 0, 8);

  //Reading the file named "data.txt".
  //ios:in means that the data is to be read "in" from the file
  fstream file;
  file.open("data.txt",ios::in);

  // declaring a variable value to store the values in each line of
     the data.txt file.
  double value;

  // while loop is used when we are not aware how many times the
    loop runs.
  while(1)
  {
    // >> operator means that each line of file is stored in
    variable value as the loop proceeds
    file >> value;
    // filling the histogram, using the values stored in the file
    hist->Fill(value);
    if(file.eof()) break;
  }

  // Closing the file as the loop terminates
  file.close();

  // Setting up the labels for the Axes
  hist->GetXaxis()->SetTitle("X_label");
  hist->GetYaxis()->SetTitle("Y_label");

  // Creating a canvas object c1 for our graph to plot on
  TCanvas *c1 = new TCanvas();

  // Drawing the final histogram by "telling" histogram object to "
    draw" on canvas.
  hist->Draw();

}
```

Listing 1: Drawing a Histogram using Root

2.2 Generating random numbers

We will use the `TRandom2` random number generator to generate 10,000 random numbers. We will then use these random numbers to fill in the histogram, for visualization purposes. By default, the numbers generated are uniformly distributed.

```
void rand_unif()
{
  // TRandom2 is the random number generator "rand" is the name of
    the object
  // By default the numbers are uniformly distributed.
  TRandom2 *rand = new TRandom2();
```

```

  //Creating object hist for our histogram
  TH1F  *hist = new TH1F("hist1", "Histogram", 20, 0, 100);

  //Filling the Histogram using the 10,000 random numbers generated
  for(int i =0; i < 10000; i++)
  {
  double r  = rand->Rndm()*100;
  hist->Fill(r);
  cout << r << endl;
  }

  //Changing the axis range for histogram
  hist->GetYaxis()->SetRangeUser(0,1000);

  TCanvas *can1 = new TCanvas();
  hist->Draw();
}
```

Listing 2: Generating Random Numbers

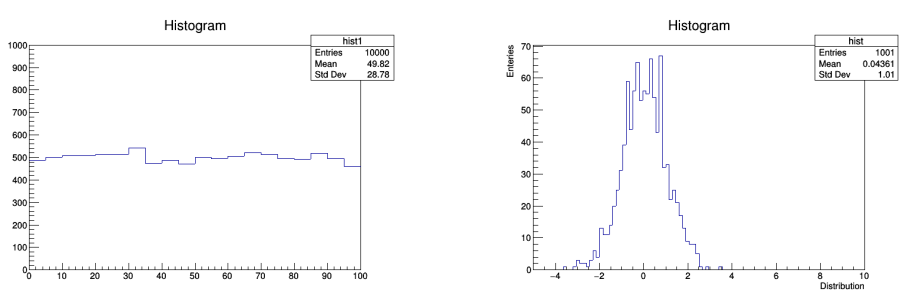
We can also create a Gaussian distribution data using the function `Gaus(mean, Variance)`, by adding the following lines to our previous code.

```
    //Loading a file "data.txt" to enter the data
  fstream file;
  //ios:out signifies that we are entering data in the file.
  file.open("data.txt",ios::out);

  //Looping through the file line by line
  for(i=0; i <1000; i++)
  {
    //Gaus(m,var) creates a random number having mean m and
    variance var.
    double r = randg->Gaus(0,1);

    //writing the value of random number in the file, line by line
    file << r << endl;
  }
```

Listing 3: Generating random numbers having a Gaussian distribution



(a) Uniform distribution

(b) Gaussian Distribution

Figure 3: Generating Random Numbers in CERN Root

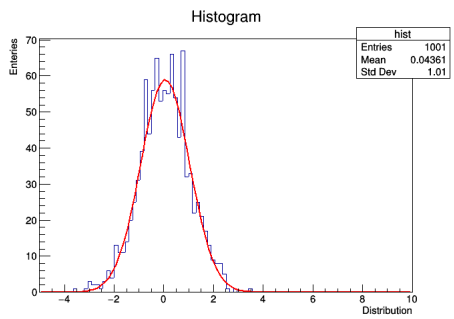


Figure 4: Fitting a Gaussian distribution to a function

2.3 Fitting a function to data

We can define a custom function or also use predefined functions of root to fit into our raw data. In the below example, we used the Gaussian distributed random data from the 2.2 example and fit using the "gaus" function. We also need to provide some initial input parameters to ensure that the fitted parameters converge.

```
  // Defining a function, "fit" which we want to fit into the data
TF1 *fit = new TF1("fit","gaus",-5,10);

// Input parameter 1: Amplitude of the Gauss
fit->SetParameter(0,70);
// Input parameter 2: Mean of the Distribution
fit->SetParameter(1,0);
// Input parameter 1:Variance of the distribution
fit->SetParameter(2,1.2);


TCanvas *c1 = new TCanvas();
hist->Draw();
// R denotes the range of the values to be fitted
hist->Fit("fit","R");

// Obtaining the final fitted parameters using the index.
double mean = fit->GetParameter(1);
double sigma = fit->GetParameter(2);
```

Listing 4: Generating random numbers having a Gaussian distribution

The output for this example is given in Figure 4

2.4 Plotting User-Defined Functions

Using the `TF1` class we can create user-defined functions in root. We need to specify the object name, the expression for the function, $f(x)$ with the parame-

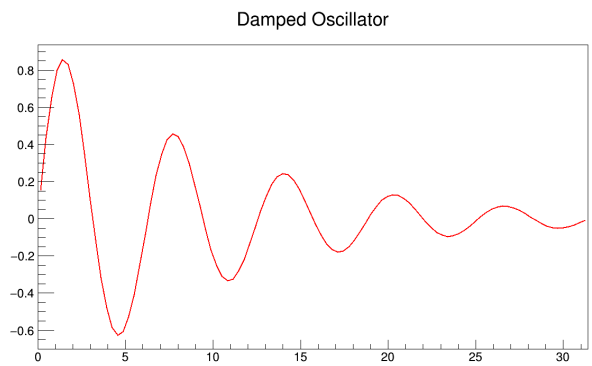


Figure 5: Visualizing a function in Root

ters in square brackets , starting from [0]. Suppose we want to plot a straight-line function $f(x) = m(x) + c$, the parameters involved are m and c. The syntax for this will be,

```
TF1 *line = new TF1("line","[0]*x + [1]",0,10);
```

In the below example we plot the function of a damped Sine wave, figure 5, commonly encountered in day-to-day life.

```
void plot_func()
{
  //Creating the object func
  TCanvas *c1 = new  TCanvas();

  // Creating our User Defined Function using the class TF1
  //Syntax: TF1("object name", f(x), x_min, x_max)
  TF1 *func = new TF1("func","exp(-[1]*x)*[0]*sin(x)",0,10*TMath::
    Pi());

  // Setting the title of the graph
  func->SetTitle("Damped Oscillator");

  // Setting the first parameter of the func, [0]
  func->SetParameter(0, 1);

  // Setting the second parameter of the func, [1]
  func->SetParameter(1, 0.1);

  func->Draw();
}
```

Listing 5: Creating User Defined functions

2.5 Filling a TTree

TTree class is designed to manage large quantities of data of similar classes in CERN root. Branches, buffers and leaves are the main objects in TTree

[9]. These objects are not written individually rather objects of the same type are collected and written in a bunch. Using TTree gives an edge over writing objects individually as TTree will compress the header objects and produces a much smaller file. Every variable can be considered to be a branch in this tree

The below example illustrates how to write an ASCII file in root specific format(TTree). The TTree is filled from a manually created file `data.txt` [10].

```
void ttree_fill()
{

  //Reading the file "data.txt" containing the data to be stored
  fstream file;
  file.open("data.txt", ios::in);

  //Declaring the variables to store values from the columns of the data file
  double x, y ;

  //Creating a new root file
  TFile *output = new TFile("data.root", "recreate");

  //Creating the TTree
  TTree *tree = new TTree("tree", "tree");

  //Defining the branches for the tree..
  //Syntax: Branch("variable_name_in_tree", &refer_variable, /Datatype_of_variable)
  tree->Branch("x", &x, "x/D");
  tree->Branch("y", &y, "y/D");

  while(1)
  {
    file >> x >> y;

    if(file.eof()) break;

    cout << x << " " << y << endl;

    //Filling the tree with the variables
    tree->Fill();
  }

  output->Write();

  output->Close();

  file.close();
}
```

Listing 6: Filling a TTree

After filling the TTree, we can run the root file and then view the root file using `TBrowser` command. Figure 6 shows the `TBrowser` window, x and y are the branches containing the histogram. The histograms can be viewed by clicking the corresponding branches.

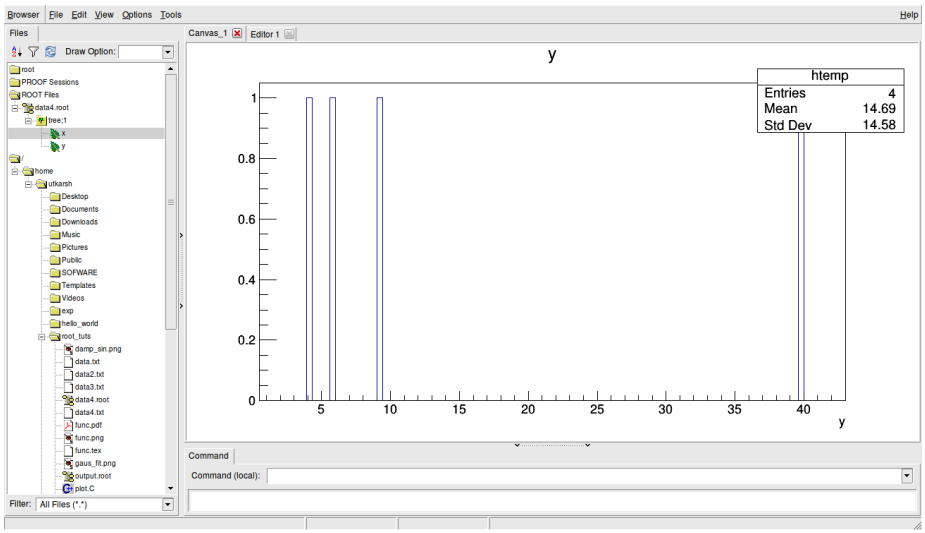


Figure 6: Viewing the TTree using TBrowser

3 CMS J/Ψ Measurement

We performed an experiment [11] wherein we used the measurements provided for the J/Ψ decay with the `ISpy Online` event viewer of CERN. We rated the candidates for J/Ψ by looking at the muon trajectories.

3.1 Theory

3.1.1 CERN-CMS

The CERN Compact Muon Solenoid (CMS)[12] is a general purpose detector at the Large Hadron Collider (LHC). It is involved in a set of diverse physical experiments ranging from searching for the dark matter to the study of the Standard Model. The CERN CMS experiment detector is built around a huge solenoid magnet, taking the form of a cylindrical superconducting coil. It generates a magnetic field of 4 Tesla.

3.1.2 J/Ψ meson

The J/Ψ meson [13] is a flavor neutral meson made up of a charm quark-antiquark park. Bound state mesons of a charm quark-antiquark are known as *Charmoniums*. It has a spin +1 and rest mass energy of 3.0969 GeV/c^2. The J/Ψ, is charge neutral as it is composed of a quark and anti-quark having equal and opposite charges.

The J/Ψ had an extremely long decay time, about 1000 times more than the normal decay time. The OZI (1.11) rule suppresses the hadronic decay modes, increasing the lifetime of the particle and giving a very narrow decay width. Due

to its well-known peak width and particle mass, it is often used as a *calibrator* in experiments.

The J/Ψ meson was first discovered by Burton Richter(SLAC) and Samuel Ting(Brookhaven National Laboratory) each working independently. The discovery of the J/Ψ meson shed light on the existence of a fourth quark, apart from the three quarks(up, down and strange) those predicted by the early models. The fourth quark is now what we know as Charm(c) quark.

The extreme narrowness of the ψ and $\psi*$ compared to the other meson resonances indicated that it is impossible to describe them in terms of the up, down, or the strange quarks. J/Ψ mesons were said to possess a vector combination of the charm quark-antiquark pair, charmonium($c\bar{c}$).

A few years ago, Glashow, Iliopoulos, and Maiani had already postulated that flavor-changing decay would be suppressed if there existed a fourth quark, Charm(c) that could be paired with Strange(s) quark.

The major decay pathways for J/Ψ are [14],

$$
\begin{aligned}
c\bar{c} &\longrightarrow 3\,g \\
c\bar{c} &\longrightarrow \gamma + 2\,g \\
c\bar{c} &\longrightarrow \gamma \\
\gamma &\longrightarrow \text{hadrons} \\
\gamma &\longrightarrow e^+ + e^- \\
\gamma &\longrightarrow \mu^+ + \mu^-
\end{aligned}
$$

The main significance of the J/Ψ were,

1. The J/Ψ meson discovery proved that the quark model is a reliable theory of nature, before its discovery the quark model was just a mathematical tool.

2. It confirmed the existence of a fourth quark, Charm.

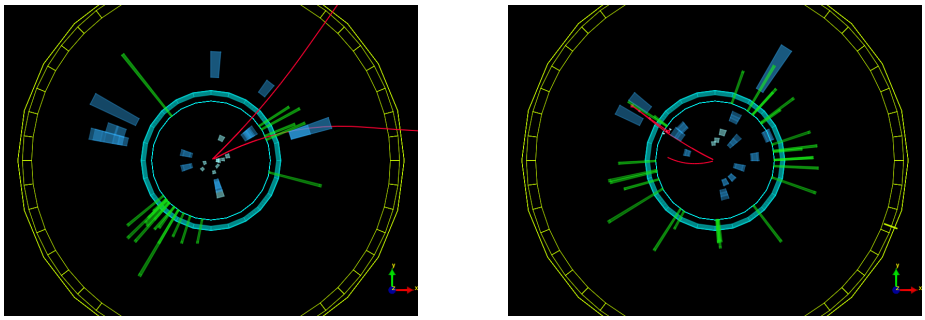
3. Due to its well-known peak width and particle mass, it is often used as a *calibrator* in experiments [15].

3.1.3 Test for J/Ψ decay

As we have seen from the above decay paths, that a J/Ψ meson decays to produce a dimuon having opposite charges. The two muons follow different tracks, which we will analyze using the event display software to find if the decay channel is J/Ψ or not.

There are two tests to determine if the decay is J/Ψ or not,

1. Test 1: The first requirement for an event to be a possible J/Ψ candidate is that the two muon tracks should not have the same directions. If the



(a) A possible J/Ψ event (b) Not a J/Ψ event

Figure 7: Muon Tracks of two potential J/Ψ events

two muons have the same direction then it means that they have the same charge and thus have been bent in the same direction by the solenoid magnet. This test is also known as the 'charge test'. Refer figure 7

2. Test 2: If the muon tracks pass the first test then we will have to rate the likelihood of each candidate for the J/Ψ event by rating the tracks individually.

3.1.4 Elements of a J/Ψ candidate event

The main elements of a J/Ψ event are,

1. ECAL Barrel: It is the Central electromagnetic calorimeter. It measures the energy of electrons and photons
2. HCAL Outer: It is the final HCAL layer just outside the solenoid magnet.
3. Hadronic Jets: They are the collimated groups of particles that come from strong force energy conversion to matter.
4. Drift Tubes (DT): They are the main large central muon chambers outside the solenoid and HCAL Outer using ionization of gas to record muon hits
5. DT Rec. Hits: They are the muon hits in Drift Tubes (barrel).
6. RPC Rec. Hits: They are the muon hits in Drift Tubes (barrel)
7. Global Muons: "Global Muon" refers to the whole track of muon.
8. Tracker muons: The "tracker" and "standalone" muon are the tracks of muon in various parts of the CMS.

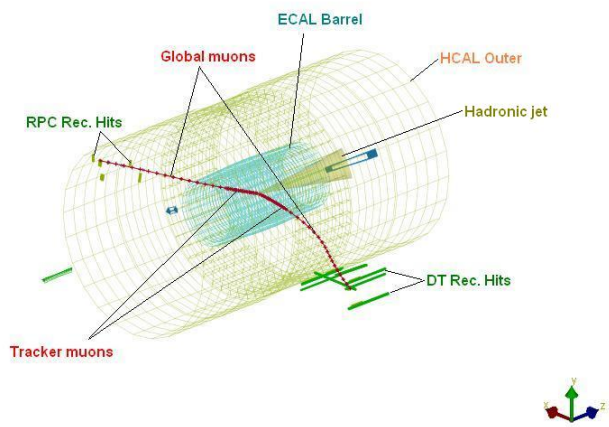


Figure 8: Elements of a J/Ψ candidate event
Figure taken from [16]

3.2 Installation and Setup

iSpy is a powerful and lightweight event display used for the CMS experiment at the LHC. It reads in an ig file which contains event and geometry information(Energy and momentum). For our experiment, we will be using `iSpy-online` event viewer.

The major steps for installation and setting up the experiment involves,

1. Visit `iSpy-online` event viewer. This is a browser based event viewer.
2. We then proceed to Collections $\longrightarrow$ J/Psi $\longrightarrow$ dimuon-j/Psi $\longrightarrow$ Open the .ig file $\longrightarrow$ Select the first Event $\longrightarrow$ Choose Load.

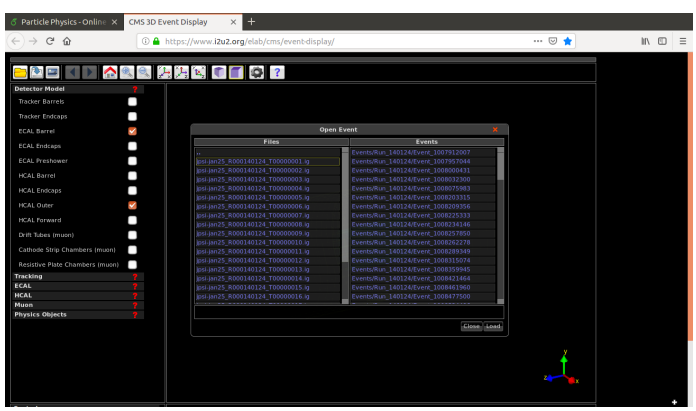


Figure 9: Loading the events

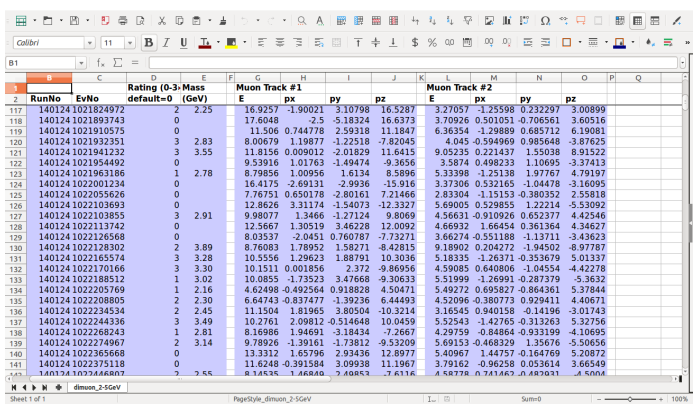
3. After loading the view we have to ensure that, **Tracks (reco)** checkbox should be off. The switch to the **x-y** view.

Figure 10: First button: XY View

4. In the x-y view, we can zoom in to see the muon tracks(red) clearly. Based on the tests [3.1.3], we rate the likelihood of being a possible J/Ψ candidate by entering a number 0,1,2,3 in the spreadsheet.

Figure 11: Buttons to adjust Zoom

5. As we enter the rating for an event, the mass for that event(in GeV), is automatically computed and entered in the table.



1						Muon Track #1					Muon Track #2				
2	RunNo	EvNo	Rating (0-3 default=0	Mass (GeV)		E	px	py	pz		E	px	py	pz	
117	140124	1021824972	2	2.25		16.9257	-1.90021	3.10798	16.5287		3.27057	-1.25598	0.232297	3.00899	
118	140124	1021893743	0			17.6048	-2.5	-5.18324	16.6373		3.70926	0.501051	-0.706561	3.60516	
119	140124	1021910575	0			11.506	0.744778	2.59318	11.1847		6.36354	-1.29889	0.685712	6.19081	
120	140124	1021932351	3	2.83		8.00679	1.19877	-1.22518	-7.82045		4.045	-0.594969	0.985648	-3.87625	
121	140124	1021941232	3	3.55		11.8156	0.009012	-2.01829	11.6415		9.05235	0.221437	1.55038	8.91522	
122	140124	1021954492	0			9.53916	1.01763	-1.49474	-9.3656		3.5874	0.498233	1.10695	-3.37413	
123	140124	1021963186	1	2.78		8.79856	1.00956	1.6134	8.5896		5.33398	-1.25138	1.97767	4.79197	
124	140124	1022001234	0			16.4175	-2.69131	-2.9936	-15.916		3.37306	0.532165	-1.04478	-3.16095	
125	140124	1022055626	0			7.76751	0.650178	-2.80161	7.21466		2.83304	-1.15153	-0.380352	2.55818	
126	140124	1022103693	0			12.8626	3.31174	-1.54073	-12.3327		5.69005	0.529855	1.22214	-5.53092	
127	140124	1022103855	3	2.91		9.98077	1.3466	-1.27124	9.8069		4.56631	-0.910926	0.652377	4.42546	
128	140124	1022113742	0			12.5667	1.30519	3.46228	12.0092		4.66932	1.66454	0.361364	4.34627	
129	140124	1022126568	0			8.03537	-2.0451	0.760787	-7.73271		3.66274	-0.551188	-1.13711	-3.43623	
130	140124	1022128302	2	3.89		8.76083	1.78952	1.58271	-8.42815		9.18902	0.204272	-1.94502	-8.97787	
131	140124	1022165574	3	3.28		10.5556	1.29623	1.88791	10.3036		5.18335	-1.26371	-0.353679	5.01337	
132	140124	1022170166	3	3.30		10.1511	0.001856	2.372	-9.86956		4.59085	0.640806	-1.04554	-4.42278	
133	140124	1022188512	1	3.02		10.0855	-1.73523	3.47668	-9.30633		5.51999	-1.26991	-0.287379	-5.3632	
134	140124	1022205769	1	2.16		4.62498	-0.492564	0.918828	4.50471		5.49272	0.695827	-0.864361	5.37844	
135	140124	1022208805	2	2.30		6.64743	-0.837477	-1.39236	6.44493		4.52096	-0.380773	0.929411	4.40671	
136	140124	1022234534	2	2.45		11.1504	1.81965	3.80504	-10.3214		3.16545	0.940158	-0.14196	-3.01743	
137	140124	1022244336	3	3.49		10.2761	2.09812	-0.514648	10.0459		5.52543	-1.42765	-0.313263	5.32756	
138	140124	1022268243	1	2.81		8.16986	1.94691	-3.18434	-7.2667		4.29759	-0.84864	-0.933199	-4.10695	
139	140124	1022274967	2	3.14		9.78926	-1.39161	-1.73812	-9.53209		5.69153	-0.468329	1.35676	-5.50656	
140	140124	1022365668	0			13.3312	1.65796	2.93436	12.8977		5.40967	1.44757	-0.164769	5.20872	
141	140124	1022375118	0			11.6248	-0.391584	3.09938	11.1967		3.79162	-0.96258	0.053614	3.66549	

Figure 12: Data Spreadsheet to enter the values

6. We proceed to the next event by clicking the **right arrow** on top of the screen. We rate all the events similarly,

3.3 Measurements

The Measurement for the experiment consists of the mass of the dimuon events, and the rating for the likelihood of the decay via J/Ψ meson. The rating scheme:0 [No Chance],[1 Poor],[2 Fair],[3 Good]. The total number of events to be analyzed was 2000.

The measurements for the first 100 events are given in the Table3.

3.4 Summary

Using, the mass values of the events we plotted a histogram, depicting the mass values of the muons(in GeV) and number of the events. The histogram was plotted by using CERN Root. The values plotted in the histogram are those for which the rating was 2 or 3.

The histogram shows a sharp peak at around 3.1 GeV. This corresponds to the rest mass energy of the J/Ψ meson.

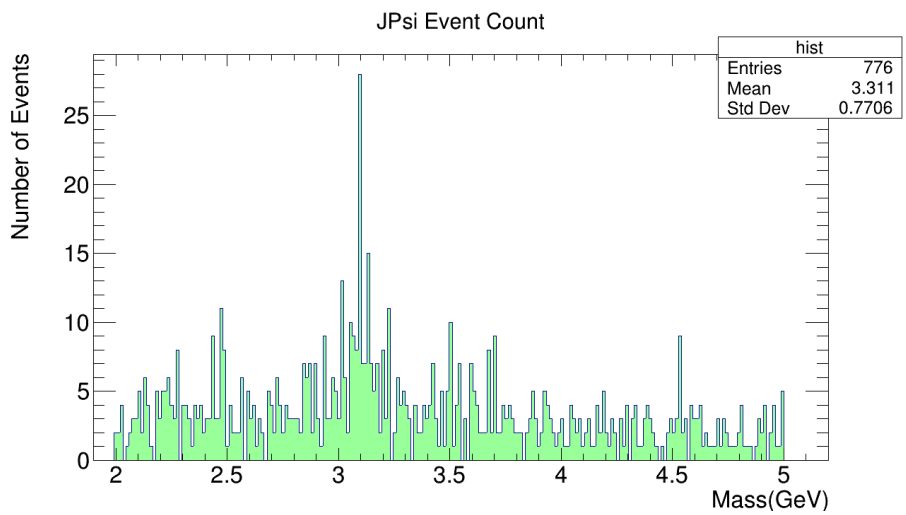


Figure 13: Histogram showing the Event Count and the Muon Mass(Gev)

We have so far outlined the basic concepts in Particle Physics. The various examples in CERN ROOT equipped us with a sound understanding of the working knowledge of data analysis and data management in Particle Physics. We have also rated the J/Ψ dimuon tracks and performed the data analysis on the data. The mass distribution plot shows the mass of the J/Ψ meson to be around 3.1 GeV, indicated by a sharp peak. The CERN masterclass provided us a great opportunity for working with real data acquired by CERN CMS. It also gave a feel of particle physics experiments and research.

4 Paper Review: *Inclusive J/Ψ production at midrapidity in pp collisions at $\sqrt{s} = 13$ TeV*

We reviewed a paper[17] by the European Organisation for Nuclear Research and ALICE collaboration related to the calculations of inclusive J/Ψ cross section. The major learnings through the paper are summarised in this section.

4.1 Basic Terms used in paper

1. Rapidity(y): Rapidity is a measure of relativistic velocity it is given by $tan^{-1}\frac{v}{c}$. The rapidity can also be obtained from energy and momentum as,
$$y = \ln \frac{E + |p|\, c}{mc^2}$$

2. Pseudorapidity(η): is a spatial coordinate, that defines the angle of a particle relative to the beam axis.
$$\eta = arctanh\Big(\frac{\rho_L}{|p|}\Big)$$

3. Total COM Energy($\sqrt{s}$): Energy of two protons colliding in a rest frame of the combined Centre of Mass of the two moving protons, such that the two protons are moving at the same speed.

4. Time Projection Chamber(TPC): In physics, the time projection chamber is a type of particle detector that uses a combination of electric and magnetic fields together with a sensitive volume of gas to perform 3-D reconstruction of particle trajectory.

5. Calorimeter: In particle, physics a calorimeter is an instrument to measure the energy of the particles. Most particles enter the calorimeter, initiate a particle shower and the particle energy is deposited in the calorimeter.

6. Electromagnetic Calorimeter: It is specifically designed to measure the energy of the particles that primarily interact via EM radiations,

7. Trigger: A system that uses a criteria to quickly decide which events in a particle detector to keep. It is used when only a fraction of total events can be recorded.

8. Spectrum Hardening : increase in the average energy of the spectrum, it may be by filtering the low energy particles.

9. Color Glass Condensate(CGC): It is a state of matter obtained when nuclei collide with the speed of light. At, the gluon recombination and splitting equilibrium, new and universal properties of hadronic matter are observed. This is CGC.

10. Branching Ratio(BR): BR for a decay is the number of particles that decay through a particular decay mode to the total number of particles that decay.

4.2 Introduction

Production of quarkonium in interactions of the hadrons with nucleus is a handy tool for understanding hadronization, in QCD.

Factorization Theorem: The J/Ψ production can be factored into Short Distance Part: $c\bar{c}$ production and Long Distance Part: Subsequent formation of a bound state.

Some of the various approaches used to find the production cross section are Non Relativistic QCD, next to leading order calculations, k_T factorization approach, Color Evaporation Model.

The main objectives of this paper are computing the integrated (ρ_t, y), differential (ρ_t) and differential (y) production cross sections of the inclusive J/Ψ production from pp collisions at midrapidity at $\sqrt{s} = 13$ TeV.

4.3 Instrumentation and Data Sample

Inner Tracking System and Time Projection Chamber are used for the track reconstruction. They are placed in a uniform magnetic field of 0.5 T along the beam. The TPC is also used in electron identification, by measuring the specific energy loss ($\frac{dE}{dx}$). The Electromagnetic(ECal) and the Di-Jet calorimeter are used as the trigger and electron identification.

The EMCal trigger is used in the measurement of production cross sections in range 15 GeV/c $< \rho_t <$ 40GeV/c. The minimum bias trigger is used to for 0 GeV/c $< \rho_t <$ 15 GeV/c. The EMCal is a two-level trigger and requires that at least one of the charge sums of the adjacent tower is $>$ 9 GeV. The integrated luminosity for the triggers is given as

$$L_{int} = \frac{N_{MB}}{\sigma_{MB}} \times \frac{ds_{trig}}{ds_{MB}} \times \frac{LT_{trig}}{LT_{MB}}$$

4.4 J/Ψ Reconstruction

The selected electron tracks are required to be under central barrel acceptance and also have a minimum transverse direction momentum ρ_t of 1 GeV/c to suppress the background. Distance of closest approach based selection was employed to reject tracks from weak decays. For the minimum bias trigger, electrons are identified using $\frac{dE}{dx}$ in the TPC. For the EMCal trigger, electrons are identified by $0.8 < \frac{E}{p} < 1.3$ and also by $\frac{dE}{dx}$ in TPC. Secondary electrons are also rejected using hits on Silicon pixel detectors.

The reconstructed J/Ψ is obtained from the invariant mass distributions of opposite sign pairs, from selected electron tracks within the same event. The two types of backgrounds are due to uncorrelated(combinatorial) and correlated processes. In MB analysis, the combinatorial background is removed from

the invariant mass distribution (through event mixing) and fitted with a two-component function, one to obtain the correlated background and the other for the J/Ψ signal. In ECal, Monte Carlo fitting is done to obtain the signal, the background is obtained by fitting a third-order polynomial.

4.5 Systematic Uncertainties

The main sources of systematic uncertainties include -

1. TPC-ITS matching uncertainty: is measured by the residual difference between data and Monte Carlo simulations for the matching. The uncertainty in track quality is obtained by repeating the analysis by varying the selection criteria and taking RMS of the resulting distributions.

2. Electron Identification Uncertainty: estimated by data driven procedure using a sample of identified electrons from tagged photon conversions.

3. Uncertainty on EMCal trigger: measured by varying random noise

4. Uncertainty in input kinematic distributions: reduced using an iterative procedure

5. Uncertainty in branching ratios and integrated luminosity(L_{int}). : Global Systematic uncertainty

4.6 Results

The double differential J/Ψ production cross section is calculated using,

$$\frac{d^2\sigma_{J\Psi}}{dyd\rho_t} = \frac{N_{J\Psi}}{BR(J\Psi \to e^+e^-)\cdot < A \times \epsilon > \cdot \Delta y \cdot \Delta\rho_t \cdot L_{int}}$$

A power-law function is fit into the distribution. The ratio of forward rapidity and mid-rapidity data points shows a monotonic decrease at high values of ρ_t showing that the J/Ψ distribution is harder at mid-rapidity. The monotonic drop in the ratio of observations at 5.02 TeV and 13 TeV shows the spectrum hardening [8] with higher collision energies.

The measured J/Ψ inclusive cross section is also compared with many other phenomenological calculations both from prompt and non prompt decays. Measured inclusive production cross section is compared with the sum of the predictions for prompt(from heavier charmonium states) and non-prompt(decay of beauty hadrons) J/Ψ calculations. The improved color evaporation model(ICEM) and Non-Relativistic QCD + Color Glass Condensate models provide predictions that agree highly with data at low transverse momentum. The earlier co linear approaches tend to diverge as ρ_t tends to 0. The integrated $J\Psi$ production cross section comes out to be,

$$\frac{d\sigma_{J\Psi}}{dy} = 8.94 \pm 0.24(\text{stat}) \pm 0.48(\text{syst}) \pm 0.45(\text{lumi}) \pm 0.05(\text{BR})\ \mu\text{b}$$

where, stat.: total statistical uncertainty; syst: total uncertainty from systematic sources; lumi: global systematic uncertainty in integrated luminosity; BR: global systematic uncertainty in the branching ratio

4.7 Conclusions

The integrated cross section vs collision plot(given in this paper) shows that the cross sections are seen to logarithmic increase with the energy. The integrated cross section vs rapidity plot shows that no rapidity dependence in mid rapidity range, however a steep decrease is observed with forward rapidity.

5 Summary and Conclusion

The mass distribution plot of the dimuon showed a sharp peak at 3.1 GeV. Thus, the mass of the J/Ψ comes out to be 3.1 GeV or 3100 MeV. The value reported by the Particle Data Group(PDG) [18] is 3096.900 ± 0.006 MeV. The value obtained through the masterclass experiment comes out to be in close agreement with the PDG value. The dimuon tracks were rated on the basis that if the two tracks curved in the same direction it means essentially they have similar sign charges, hence they could not be a result of J/Ψ decay. On the other hand, if the tracks were going in opposite directions, it meant that the muons were oppositely charged, such tracks were given a high likelihood of being due to a possible J/Ψ decay. The property of J/Ψ meson being charge-neutral was being exploited here.

This project greatly helped me to build a fundamental understanding of the basic concepts involved in Particle Physics like Particle Classification, Fundamental Interactions, Klein Gordon Equation, Symmetry, and Conservation Laws of Particle Physics. Apart, from developing a theoretical understanding of introductory Particle Physics, the introduction to CERN Root uncovered a whole new arena of data analysis and plotting tools especially curated for Particle Physics. The virtual experiment to classify the possible J/Ψ candidates was very insightful to get aware of the real-life experiments taking place at CERN-CMS. It also provided great knowledge about managing and visualization of a large amount of data, that particle physicists face.

The experiment also made me appreciate the importance of the J/Ψ meson and how it led to the formulation of a whole new quantum number, charm. It also accelerated the whole new avenue of experimental Particle Physics research and showed that how theoretical and experimental models are intertwined with each other. The importance of this discovery in bringing about rapid changes in particle physics is quite evident from the name, November Revolution.

The paper review made me aware of tons of terms and techniques used in various experiments. I was also able to understand and appreciate the various challenges that are being faced in data acquisition, processing, and management. The methods to curb these challenges were also emphasized. The paper helped me understand various recent aspects in particle physics usually not found in contemporary literature.

I believe that this summer project will greatly benefit me in doing a further study of Particle Physics and also help me to take up complex topics with ease. The various computational tools used in this project have enabled me to get a taste of real research. I will look forward to expanding my knowledge of using CERN Root to use various classes like TClone, TChain, and other advanced concepts.

6 Appendix

Table 3: Measurements of the events(100 events)

Rating (0-3) default=0	Mass (GeV)
3	4.52
1	2.73
2	3.07
0	
1	2.16
1	2.11
0	
0	
0	
1	2.83
3	3.27
1	2.79
0	
0	
2	2.45
2	2.03
3	4.19
3	4.12
2	3.61
3	2.57
3	3.70
0	
1	2.51
0	
0	
1	2.57
2	3.48
3	3.21
2	2.59
2	3.01
1	2.50
3	3.44
3	3.02

Rating (0-3) default=0	Mass (GeV)
2	3.87
0	
2	2.27
3	2.83
1	2.62
1	3.45
1	2.72
1	2.63
2	2.49
3	2.10
1	2.07
1	2.04
3	4.21
2	3.57
0	
1	2.58
2	3.10
3	2.82
2	2.21
2	3.09
1	2.91
1	2.96
3	3.07
3	4.58
2	3.16
0	
1	2.80
2	2.49
0	
1	2.99
2	3.10
2	4.94
2	3.62

Rating (0-3) default=0	Mass (GeV)
2	3.78
0	
1	3.17
2	3.47
0	
1	2.58
3	3.93
2	2.13
3	3.11
0	
1	2.84
0	
2	2.10
3	3.03
1	2.04
1	3.22
1	3.77
3	3.32
3	3.18
2	3.06
2	2.90
0	
0	
3	4.74
3	3.13
1	2.03
3	3.68
3	4.50
3	2.27
2	4.52
0	
1	3.39
2	3.55
1	2.59

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